

Heidi Visualization of R-tree Structures over High Dimensional Data

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Abstract. High dimensional index structures are used to efficiently answer range queries in large databases. Visualization of such index structures helps in: (a) *visualization of the data set* in a hierarchical format of the index structure, (b) *“explorative querying”* on the data set, similar to explorative browsing on the web, (c) *index structure diagnostics*: visualizing the structure along with its performance statistics enables the user to make changes to structure for better performance. To the best of our knowledge, there is no such visualization for high dimensional index structures.

In this paper, we present Heidi Visualization of R-Trees. Heidi Visualization is a two-dimensional visualization of high dimensional data and displays “close-ness” of points across various subspaces. The Heidi Visualization could be applied over any other index structure too. In this paper, we demonstrate Heidi Visualization over the R-Tree and R*-Tree.

1 Introduction

Data Visualization is a crucial task in decision making [11]. Looking at the large volumes of data is practically not feasible. Data visualization needs to be (a) *compact*: to ensure all the data is visualized in one screen shot ideally and (b) *effective*: to ensure the information obtained from the data is comprehensible to the user.

Processing large volumes of data brings forth the efficiency and performance issues. Building index structures to access data quickly is a popular and efficient solution for fast data access. The R-Tree structure and its variants are popularly used for indexing numerical data. R-Trees can be built using various splitting rules. These index structures, apart from pruning the nearest neighbor search space, also have some inherent data analytics that could be computed to help understand the high dimensional data better; for instance, different sizes of MBRs (size measured as number of points) help estimate on the density of the MBRs based on their respective ranges. A visualization mechanism of the index structures helps user observe the changes in the index structure when various splitting rules and branching factors are applied to generate them. Heidi Visualization [13] is a 2-D visualization of a high dimensional data set which emphasizes the clusters present in the data and the various subspace overlaps among the clusters. The overlap information is obtained by computing the k nearest neighbors across various subspaces (subsets of dimensions).

In this paper, we present R-Tree/R*-Tree index structure data visualization and data analytics for high dimensional data using Heidi Visualization system. To the best of the our knowledge, there has been little work [9] done in visualizing R-Trees for high dimensional data.

1.1 Motivation

Spatial databases and geographic information systems have been established as a mature field. In multimedia databases (like image, music and voice), R-Trees [10] are a necessary tool for structured data storage and for effective retrieval upon queries.

Benefits of Index Structure Visualization

Visualization of the index structures for high dimensional data has various benefits:

1. *Understanding* the broad hierarchical index structure defines over the data set in high dimensional space.
2. Ability to visualize these structures brings forth the possibilities of *evaluating* the structure and making required amendments to the structure to *improve* its performance and efficiency.
3. The structure of the R-Tree and its variants are a reflection of the *data distribution* of points. MBB statistics like number of MBBs, number of points in MBBs and density of the MBBs can help to bring forth the dense regions vs. the sparse regions in the data set. High-dimensional data has been indexed by mainly focusing on the observation that real data in high-dimensional space are highly correlated and clustered [4], and therefore the data occupies only some sub-spaces of the high-dimensional space.
4. High dimensional data suffers from the curse of dimensionality and is sparse. There are more chances of finding dense sets of points in certain lower subspaces than the original space. Lower subspaces might have better structure in terms of better groups of close points (points that have lesser distances among themselves). Hence, the visualization should be able to show these subspace overlaps.
5. Sparsity in High Dimensional Spaces: Though R-Tree answers high dimensional range and nearest neighbor queries, there are instances when the points that satisfy a given query might have an overlap with another set of points in a particular subspace. We need to visualize such interesting sets.

In this paper, we enhance Heidi system to display not only clusters but also hierarchical index structures like R-Tree and R*-Tree. We also provide an alternative Heidi image providing prominent subspace information.

The organization of the paper is as follows: Section 2 covers a brief history of R-Tree, its variant R*-Tree, Heidi Visualization and the related work. Section 3 states the problem we are addressing and presents Heidi Visualization for index structures followed by an example. Section 4 presents the results and Section 5 concludes the paper.

2 Background

2.1 R-Tree

A R-tree has a B+-tree [7] like structure which stores multi-dimensional rectangles as complete objects without clipping or transforming them to higher dimensional points. A multi-dimensional rectangle is referred to as a Minimum

Bounding Box (MBB¹) and it bounds a set of objects that are located within its boundaries. The MBB for a point set in d dimensions is defined as the box with the smallest measure (area, volume, or hyper-volume in higher dimensions) within which all the points lie.

R-Tree structure is a hierarchical structure defined over the points; points grouped into MBBs and MBBs are again grouped into larger MBBs. A non-leaf node contains entries of form (cl, MBB) , where cl is the list of addresses of the child nodes and MBB encloses the rectangles of the child nodes *minimally*. A leaf node contains entries of form (pl, MBB) , where pl is the list of points present in the corresponding MBB.

R-Tree properties

1. Let M be the maximum number of points that will fit in a node and m be the minimum number of points in a node. Every leaf node aims to have at least m and at most M points.
2. When the leaf node found can not accommodate any more points then the node is split using one of the splitting techniques.

Splitting Techniques

R-Trees differ in how they perform splits during insertion by considering different minimization criterion. Basic splitting techniques that were proposed in the very first work on RTree by Guttman [10] were Linear Split, Quadratic Split and Exponential Split. For different splitting techniques different subsets of points are grouped in a MBB. In this paper, we implemented the Linear Split and in this technique, two farthest objects are chosen as seeds. Each remaining object is assigned to the closest seed.

2.2 R*-Tree

R*-Trees [3] were proposed in 1990 and are widely used. Objective of the R*-Trees is to minimize the area covered by MBB, overlap between MBBs, MBB margins and storage utilization. When we insert a new point in a leaf node that is already full, the split dimension with minimum perimeter is selected. When split dimension is selected, split algorithm sorts the points on this dimension for all possible divisions. For each axis, the overlap of $(M - 2m + 2)$ divisions is calculated (M is the maximum number of points that will fit in a node and m is the minimum number of points that will fit in a node). The final division is the one that has minimum overlap between resulting nodes.

2.3 Heidi Visualization: A Brief Overview

Heidi [13] is a high dimensional data visualization system that captures and conceptually visualizes the close-ness of points across various subspaces of the dimensions; thus, helping to understand the data. The core concept behind Heidi is based on prominent patterns within the nearest neighbor relations between pairs of points across the subspaces. This representation acts is like an X-ray of the data set, highlighting the relevant patterns in various subspaces. It enables the user to visualize and reason about the close-ness of data points across all the subspaces, by displaying how the proximities of points change in various subspaces.

¹ For 2-D data, a MBB is referred to as Minimum Bounding Rectangle (MBR)

Given a d -dimensional data set $\mathcal{X}$ and a set of clustering results as input, Heidi system generates a 2-D matrix (called Heidi Matrix) which reflects the close-ness information of every pair of points in the data set across all subspaces. The 2-D matrix is represented and used as a color image and gives insight into (i) how the clusters are placed with respect to each other, (ii) characteristics of placement of points within a cluster in all the subspaces and (iii) characteristics of overlapping clusters in various subspaces. The *subspaces* are non-empty subsets of the dimensions of data set $\mathcal{X}$. The notation followed in this paper is given in Table 1.

$\mathcal{X}$	data set
n	size of the data set
$\mathcal{D}$	set of dimensions
$\mathcal{P}$	set of all possible subspaces
$\delta_{\mathcal{S}}(p, q)$	distance between two points p, q in subspace $\mathcal{S}$
MBR	Minimum Bounding Rectangle (for $ \mathcal{D} = 2$)
MBB	Minimum Bounding Box (for $ \mathcal{D} > 2$)

Table 1. Terms and Notation

k -Nearest Neighbors (k NN)

The k -th nearest neighbor of a point $kNN_{\mathcal{S}}(p)$ forms a set of all points that are at most k -th nearest to point p in subspace $\mathcal{S}$.

Close-ness between points

k NN relationships are used to define the close-ness between a pair of points. A point-pair (p, q) is considered “close” in a subspace if p considers q as one of its k NNs in that subspace. k NN relationship for a point pair is considered across all the subspaces.

Heidi Matrix

Heidi Matrix is 2-D $n \times n$ grid of bit vectors of size equal to the number of possible subspaces. The basic unit of abstraction in Heidi matrix is a bit. A bit (p, q, S_i) is set to 1, if point p is k -nearest to point q in subspace S_i . The set of bits corresponding to all possible subspaces together constitute the bit vector which is the value of the cell (p, q) of the Heidi matrix. Groups of adjacent cells belonging to a single cluster either along the rows or columns form the blocks. At a higher level, a block B_{ij} ($1 \leq i, j \leq c$, c = number of clusters) represents “close-ness” relations existing between the points belonging to cluster C_i and cluster C_j . Each block B_{ij} constitutes a set of $|C_i| \times |C_j|$ cells ($|C_i|$ and $|C_j|$ are number of points in the clusters C_i and C_j respectively), each cell of the matrix (p, q) represents whether q is a k NN of p across various subspaces.

Heidi Visualization

When the points are ordered cluster-wise along the rows and columns of the Heidi matrix, the matrix is partitioned into $c \times c$ (c representing the number of clusters) number of blocks of adjacent cells. With the Heidi Matrix computed, the integer value of bit vector is used to obtain a unique color for every possible set of subspaces. Heidi uses the RGB color model; the color intensity value of each component (i.e., R, G, B) is stored in 1 byte. Altogether, 24 bits are required to represent the color.

2.4 Related Work

R-Tree Visualization for two-dimensional data was addressed by Brabec et al. [5] and [6]. The Two-dimensional R-Tree visualization applet [6] developed, is a small part of a much more general system called VASCO (denoting Visualizing and Animating Spatial Constructs and Operations) that contains JAVA(TM) applets for a much larger set of hierarchical spatial data structures. The applet enables users to see in an animated manner how a number of basic spatial database search operations are executed for them. The spatial operations are spatial selection (i.e., a window or spatial range query) and a nearest neighbor query that enables ranking spatial objects in the order of their distance from a given query object. The results of the different splitting rules and the algorithms are visualized and animated. Here they provide a mechanism under the user's control for viewing the aggregations of bounding boxes at different levels of decomposition one level at a time, as well as combinations of several levels at a time. But the display is limited two-dimensional domain. A detailed report on this [5] does not state about how VASCO performs for very high dimensional data. R-Tree along with Parallel Co-ordinates and Star Co-ordinates was used in [9] work as a technique to visualize the hierarchical structure of data. Whenever the root node is split, a new LOD (level of details) in the hierarchy is introduced. Every internal node contains a number of children and a region which bounds all of its children. Using parallel co ordinates hyperbox is shown by plotting maximum and minimum and the area is filled between both segments. Star Co-ordinates plots all possible combinations of the minimum and maximum values in each dimension the corners of the hyperbox and fill the area by calculating the convex hull of these points and constructing its respective convex polygon. The paper mentions only within hyperbox details (number of children and region bounding it) *but we present within MBB(hyperbox) and between MBBs details as number of children, region bounding and also point-point interaction within these MBBs in various subspaces.*

3 Heidi Visualization of R-Trees

Heidi Visualization for Index Structures: Heidi visualizes high dimensional data clusters; we propose that by changing the point-ordering as per the index structure hierarchy, the visualization would reflect on the index structure characteristics. The MBBs are like the clusters having groups of points within a bounding box and hence, Heidi brings forth the various subspace overlaps between the MBBs. We customize Heidi with respect to MBBs and the hierarchical structure of the index. We generate data analytics pertaining to the MBBs and the overlaps between them across various dimensions. The hierarchy of the index structure is restored by grouping points based on the MBBs to which they belong and then ordering the MBBs with respect to the corresponding parent MBBs and so on. The point ordering techniques and MBB data analytics are mentioned below. In this paper, we choose to visualize R-Tree and R*-Tree as they are popular. Of course, the same visualization could be obtained for other index structures too.

3.1 Problem Statement

Given a d -dimensional data set $\mathcal{X}$ ($n = |\mathcal{X}|$), the *aim* is to generate a single Heidi visualization that stores comprehensive information related to

- (i) close-ness of points in various subspaces,
- (ii) MBBs of multi dimensional index structure in various subspaces,
- (iii) the hierarchical structure of the MBBs (and the R-Tree) and
- (iv) MBB overlaps in various subspaces.

Let the set of subspaces $\mathcal{P}$ considered for the R-Tree visualization be the possible single-dimensional subspaces along with the original dimensional space $\mathcal{D}$. $\mathcal{P} = \{\mathcal{D}_1, \mathcal{D}_2, \dots, \mathcal{D}_d\}$

3.2 Grouping and Ordering Points Based for Heidi Visualization on R-Tree

The order of the data points along the rows and columns of the Heidi Matrix play a significant role in visualization for giving better visualization and understanding of the internal structure of the data. With no specific point-order (say randomized), the image will not have any valuable visual information to offer.

The R-Tree structure is used in ordering the points according to MBBs and then ordering the MBBs according to their distance to their parent MBB.

The algorithm to obtain the R-Tree point ordering is given below:

1. **Building R-Tree:** For all the points in the data set, the R-tree is built. Now we insert all n points in root node. If $n > MAX$ then we split the node and repeat it recursively. Splitting is done according to linear splitting as proposed by Guttman [10]. The distance metric used is Euclidean distance of point from the origin.
2. **MBBs Ordering:** We start from the root node of the R-Tree and obtain an inorder traversal of the MBBs. Inorder traversal will return the underlying set of MBBs in order such that it groups the hierarchical information of the parent and its children together. The points belonging to the MBBs in the inorder traversal are grouped together. The MBBs and the points are further ordered within these groups on the basis of the Euclidean distance.
Suppose node N_i has children nodes $N_{i1} \dots N_{ip}$. Then centroid of MBB of N_i be denoted by $P=(p_1, p_2, \dots, p_d)$ and centroid for MBB of any children node be denoted as $P_i=(p_{i1}, p_{i2} \dots, p_{id})$.
We order the children nodes on basis of Euclidean distance between P and P_i . And the children having minimum distance value, from that we repeat the above mentioned step recursively.
3. **Points Ordering:** If the MBB is a leaf node then the points belonging to the MBB are ordered according to Euclidean distance from the centroid of the MBB. The points in each sub-tree are nearer as compared to points in another sub-tree. We maintain the hierarchical structure of tree.

After obtaining the point-order based on the index structure, the rows and columns of the Heidi Matrix are re-arranged accordingly to obtain the Heidi Visualization of R-Tree.

3.3 Implementation Details

Complexity: The complexity of computing Heidi Matrix is $O(dn^2)$. The complexity of computing ordering from R-Tree/R*-Tree index structure is $O(n)$. Space complexity is also same.

Implementation Issues: The system is developed in GNU C/C++ on a Linux system. All the images are generated in PPM image file format(.ppm). As d and n increases memory allocation is an issue for the Heidi Matrix. The time taken to compute the Heidi matrix for the examples shown in the paper ranges between 1-5 minutes. A Heidi image is obtained from Heidi matrix in few seconds time.

3.4 Data Analytics for R-Tree and R*-Tree Index Structures

Heidi highlights the pixels corresponding to pairs of points that share k NN relationship. Along with this visualization, we also obtain some statistics on the MBB-MBB overlap and the subspace in which overlap is maximal. Points within a MBB can share k NN relationship within points in another MBB in a lower subspace. From this we can find out in which subspace is this MBB important or has lesser spread as compared to others. Along with the visualization, we also provide the statistics about the overlap between MBBs and the various subspaces.

We present the following attributes: (i) Number Of MBBs, (ii) depth of the MBB in the hierarchy (d) i.e. length of the path from the root to the MBB (iii) MBB density (calculated as the fraction of points present from total number of points) and (iv) Inter MBB pattern (statistics for each subspace and in which subspace is the overlap maximum).

3.5 Explorative Querying

To execute a query, the user has to know well in advance the specific query he/she is interested in. Often with high dimensional data and with no visual aid, it is tough to compose interesting queries. In this sub-section, we demonstrate how to visualize query results in Heidi Visualization and we also present an interactive interface over Heidi Visualization using which the user can compose queries and visualize the results with just mouse-clicks on the color patterns he/she observes in the R-Tree Heidi Visualization. We term this as “explorative querying” where in the user has no specific query in mind, but after looking at the visualization and the patterns, the user tries to identify interesting sets of points satisfying a query criteria.

The most common operation with an R-Tree index is a range query, which finds all objects that a query region intersects. The problem of finding k NNs from R-trees has been introduced by Roussopoulos [12]. A branch and bound algorithm was developed to avoid the examination of entire index structure. Currently, we present an interface where the user selects the point (by clicking on the row/column of the Heidi Visualization) and enters the k value for the k -nearest neighbor query visualization and the subspace. The set of points satisfying the query are obtained and are highlighted in the Heidi Visualization. From this we can infer how optimal is the given index structure for the k NN queries based on the spread of the query result across various MBBs. If the query result falls within one MBB, then the data is structured properly for that query.

3.6 Example

Data Set

Consider a 2-dimensional data set given in Figure 1 (a). The data set has 1000 points. The points form two non-intersecting semi-circles, but the semi-circles overlap either along the x-axis or the y-axis partially.

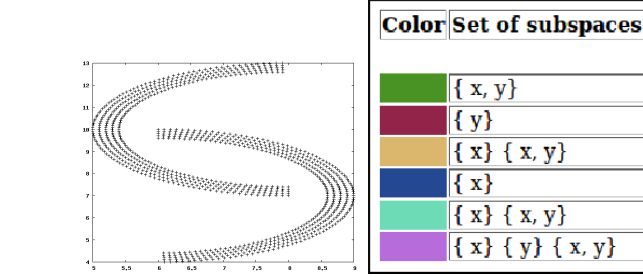


Fig. 1. (a) Semi Circle Data set, (b) Heidi Color Legend

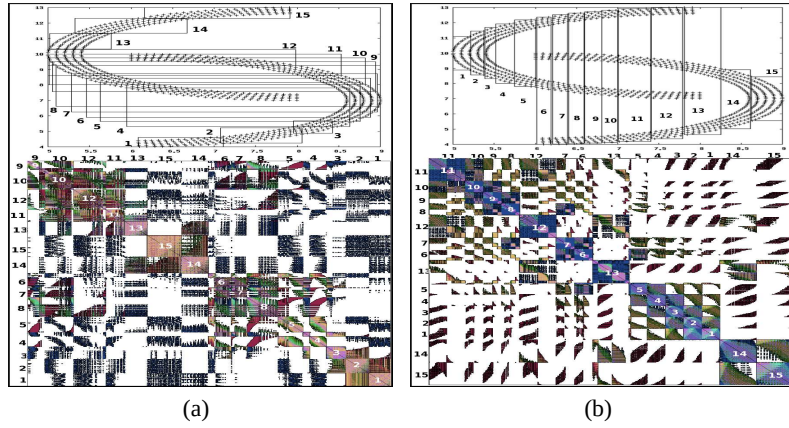


Fig. 2. R-Tree and R*-Tree Heidi Visualization for Semi-Circle Data Set:(a) Heidi Visualization of R -tree; (b) Heidi Visualization of R*-Tree

Heidi Visualization of R-Tree and R*-Tree Index Structures

The 2-D visualization of MBBs on the data set and the Heidi Visualization of the R-Tree for the data set are shown in Figure 2(a). The blocks highlighted in the 2-D R-Tree visualization denote the MBBs created in the R-Tree; it can be noticed that there is a lot of overlap among the MBBs. The Heidi Visualization displays a 1000 x 1000 pixel visualization, each pixel denoting a point pair (p, q) and the color denotes the subspaces in which they consider each other their

nearest. A total of 15 MBB leaf-nodes are present. The Heidi Visualization is also divided into 15×15 blocks. The size of these blocks depends on the number of points present in the MBBs. The blocks on the diagonal of the visualization are related to within MBB point-point interaction and the rest of blocks relate to the MBB-MBB point-point overlap in various subspaces.

In the visualization (Figure 2(a)), color present in a block denotes that the points corresponding to the related MBB block have k NN relationships in a subspace corresponding to that color. Different color patterns in the blocks signify overlaps in multiple subspaces. The color legend lists out the colors and the associated subspaces in Figure 1(b). For example, the presence of color blue, which can be noticed in many blocks in the Heidi visualization denotes that the point-pairs within these MBBs share k NN relationships in the subspace $\{x\}$. In the visualization, the blocks corresponding to the MBBs (along the diagonal) have shades of pink and purple predominantly; indicating that points within these MBBs are close to each other in subspaces $\{x\}$, $\{y\}$ and $\{x, y\}$. White color indicates that points within MBBs are not close to each other in any subspace.

Figure 2(b) displays the R*-Tree diagram over the semi circles data set and the Heidi Visualization of the same. From the 2-D diagram of the MBBs, it can be noticed that the overlap is minimized between the adjacent MBBs, though the MBBs overlap along subspace $\{y\}$. The same is reflected in the Heidi Visualization. The MBB overlaps are mostly in subspace $\{y\}$. Notice large MBB like MBB 13 which has two groups of close points (MBB-12 and MBB -14), due to which the corresponding Heidi also displays colored blocks (see at position block(MBB-13, MBB-12) and block(MBB-13, MBB-14)). MBB 15 have only one group of close points (MBB -14) and corresponding Heidi block(MBB-15, MBB-14) have shades of dusty brown, pink indicating points are close in all subspaces $\{x\}$, $\{y\}$ and $\{x, y\}$.

Comparing R-Tree and R*-Tree MBB Overlaps:

1. In the R-Tree, the MBBs overlap in both the dimensions as can be seen in Figure 2(a). Due to these overlaps, Heidi Visualization depicts presence of color patterns in the inter-MBB blocks (blocks away from the diagonal). The colors corresponding to $\{x, y\}$ space.

In the R*-Tree, the MBBs do not overlap in the original 2-D space, but they do overlap along dimension $\{y\}$. This is reflected in the corresponding Heidi visualization by the maroon color and sandy brown signifying the common subspace $\{y\}$.

2. The color pattern in the inter-MBB blocks in R-Tree's corresponding Heidi Visualization is spread across the blocks, while in the R*-Tree's Heidi Visualization, the spread is minimal and occupies only a part of the Heidi block.

It can be noticed that R* minimizes MBB-MBB overlaps in $\{x\}$ and $\{x, y\}$ subspaces better than R-Tree. Hence non-diagonal blocks in Figure 2(b) are highlighted less with dusty brown and blue shades (color pertaining to $\{x\}$ and $\{x, y\}$ subspaces) as compared to blocks in Figure 2(a) R* partitions along dimension x . Hence each MBB in Figure 2(b) along diagonal gets mostly blue color pertaining to the $\{x\}$ subspace chosen by R*-Tree.

Observing and analyzing the color patterns and their spread in the Heidi blocks gives an insight into the MBB-MBB overlap. Multiple color patterns in the same Heidi block signifies overlaps across multiple dimensions. Many inter-MBB blocks having color patterns signifies more overlap among them in various subspaces. These observations can help in analyzing which index structure is more efficient

for queries (k NN and range queries). We explain with few example queries below. Based on the type of queries (k NN and range queries), changes could be made to the structure (like merging or splitting MBBs) to enhance the performance.

Id	Den -t{y}%	d	MBB 9	MBB 10	MBB 12	MBB 11	MBB 13	MBB 15	MBB 14	MBB 6	MBB 7	MBB 8	MBB 5	MBB 4	MBB 3	MBB 2	MBB {y}
9	4.8	5	{y} 859	{y} 828	{y} 555	{y} 161	{x} 484	{x} 14	{x} 79	{y} 538	{x} 375	{x} 927	{y} 905	{x} 279	{x} 370	{x} 39	- 0
10	7.9	5	{y} 830	{y} 2152	{y} 2043	{y} 687	{x} 551	{x} 671	{x} 252	{x} 515	{y} 606	{y} 909	{x} 618	{x} 490	{x} 502	{x} 593	{x} 308
12	8.9	5	{y} 566	{y} 2258	{y} 3284	{y} 1738	{x} 526	{x} 1071	{x} 500	{x} 411	{y} 159	{y} 846	{x} 772	{x} 619	{x} 451	{x} 796	{x} 656
11	5.2	5	{x} 156	{y} 946	{y} 2061	{y} 1653	{x} 526	{x} 725	{x} 256	{x} 135	{x} 79	{x} 247	{x} 439	{x} 366	{x} 33	{x} 416	{x} 485
13	6.2	3	{x} 482	{x} 512	{x} 456	{y} 973	{y} 3422	- 0	{y} 1351	{x} 202	{x} 423	{x} 1066	{x} 66	{x} 2	- 0	- 0	- 0
15	9.4	3	{x} 10	{x} 654	{x} 1049	{x} 729	- 0	{y} 8144	{y} 1256	{x} 339	{x} 9	- 0	{x} 1120	{x} 860	{x} 66	{x} 1329	{x} 1045
14	7.4	3	{x} 89	{x} 256	{x} 507	{x} 252	{y} 965	{y} 1805	{y} 4630	{x} 701	{x} 484	{x} 442	{x} 772	{x} 381	- 0	- 0	{x} 663
6	6.3	5	{y} 549	{x} 510	{x} 397	{x} 144	{x} 240	{x} 345	{x} 691	{y} 1267	{y} 792	{y} 845	{y} 1290	{y} 888	{y} 182	{x} 435	{x} 151
7	4.3	5	{x} 372	{y} 626	{y} 178	{x} 61	{x} 429	{x} 11	{x} 489	{y} 799	{y} 808	{x} 961	{y} 479	{y} 367	{x} 21	{x} 18	{x} 14
8	8.6	4	{x} 960	{y} 928	{y} 926	{x} 208	{x} 1041	- 0	{x} 382	{x} 949	{x} 1005	{y} 2180	{y} 1397	{y} 1317	{x} 72	- 0	- 0
5	8.7	5	{y} 841	{x} 590	{x} 747	{x} 439	{x} 97	{x} 1118	{x} 782	{y} 1235	{y} 392	{y} 1433	{y} 2410	{y} 1733	{y} 423	{x} 695	{x} 744
4	6.4	5	{x} 265	{x} 472	{x} 596	{x} 369	{x} 2	{x} 865	{x} 376	{y} 837	{y} 190	{y} 1170	{y} 1582	{y} 1563	{y} 589	{x} 272	{x} 799
3	3.5	4	{x} 413	{x} 550	{x} 476	{x} 43	- 0	{x} 101	- 0	{x} 203	{x} 29	{x} 129	{y} 301	{y} 501	{y} 1156	{y} 1320	{y} 97
2	6.3	3	{x} 29	{x} 590	{x} 785	{x} 409	- 0	{x} 1341	- 0	{x} 421	{x} 14	- 0	{x} 708	{x} 268	{y} 789	{y} 3314	{y} 2086
1	6.1	3	- 0	{x} 302	{x} 657	{x} 482	- 0	{x} 1035	{x} 628	{x} 137	{x} 10	- 0	{x} 724	{x} 791	{y} 13	{y} 2560	{y} 3527

Table 2. R-Tree index analytics for 2-D Semi Circle Data Set

Data Analytics for R-Tree and R*-Tree Index Structures

Table 2 provides data analytics for R-Tree index structure over semi-circle data set. We note that MBB 15 is the most dense amongst all and is present at 5th level of tree. When we observe inter MBB overlap patterns, the (MBB-i, MBB-j) cell has two values. The first value denotes the subspace in which the MBBs overlap maximally. The second value denotes the number of pairs (x_i, x_j) share k NN relations ($x_i \in \text{MBB-i}$ and $x_j \in \text{MBB-j}$). For MBB-1, it has maximum number of pairs overlapping in subspace $\{y\}$. Notice figure 1 shows that points overlap

either in $\{x\}$ or $\{y\}$ subspace which also gets depicted in inter MBB overlaps. There is no such pair of MBBs which overlaps maximally in $\{x, y\}$ subspace. Visualizing these tables as scatterplots gives rise to Heidi Visualization at a higher coarse level for general overview about MBB overlaps. These scatter plots were generated using [1]. Figure 3 visualization is Heidi visualization at a higher level and at MBB level. It gives the overlap between MBBs. The size of the dot denotes the extent of the overlap in some subspace.

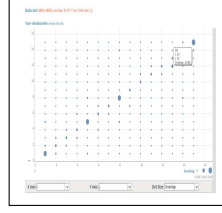


Fig. 3. Heidi Visualization at MBB level

***k*NN Query Result Visualization**

Figure 4 shows an example of a *k*NN query of point x_{10} for 10 nearest neighbours, in the semi circle data set mentioned (Figure 1) in subspace $\{0\}$. All the *k*NNs are present in 1 MBB in the Heidi Visualization of R*-Tree as shown in 4(c). For the R-Tree Visualization, the same query result is spread across more than 3 MBBs as shown in Figure 4(a). This indicates that R*-Tree's structure would need to check only 1 MBB to obtain the query result, while R-Tree (due to MBB overlaps) would need to check at least 3 MBBs.

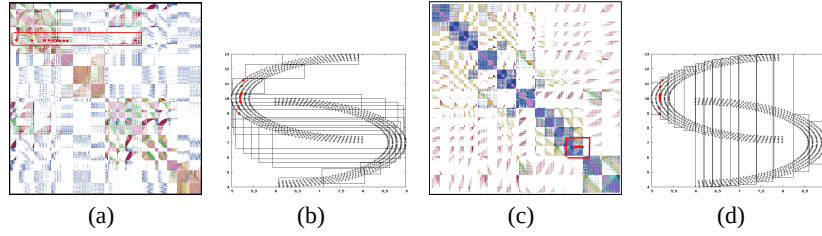


Fig. 4. Explorative Querying over Semi-Circle Data Set: (a) Query Result Over R-Tree Index Structure; (b) Query Result over data plot; (c) Query Result over R*-Tree Index Structure; (d) Query Result over data plot

3.7 Minimalistic Heidi

As P increases unique number of pixel colors also increases. Highlights in a block become difficult to be interpreted by the user. Therefore, we present minimalistic

view of Heidi visualization but which provides meaningful information. In each block we find the subspace in which overlap is maximum. Pixels of that block are highlighted with color depicting that subspace. But suppose in a block if the pixel highlights are with same colour i.e. points share kNN relationship in same subspace then we do not highlight all pixels by same color instead we let the highlights be same as it is in Heidi visualization.

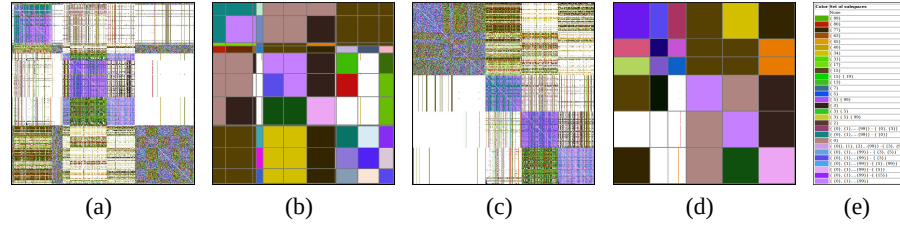


Fig. 5. R-Tree and R*-Tree Heidi Visualization over 100-d data set: (a) Heidi Visualization of R-Tree; (b) Minimalistic Heidi Visualization of R-Tree; (c) Heidi Visualization of R*-Tree; (d) Minimalistic Heidi Visualization of R*-Tree; (e) Heidi Color Legend

Example

We generated a synthetic data set of 1000 100-dimensional points using SynDeca data set generator [14]. The Heidi visualization along with the correspondin minimalistic view is shown in Figure 5. The Minimalistic Heidi Visualization of the R-Tree over 100-d dataset is displayed in Figure 5(b). It has 11 MBBs. Heidi image is divided in 11 x 11 blocks. Figure 5(d) displays the R*-Tree Index structure visualization. R* index structure has has 6 MBBs. Heidi image is divided in 6 x 6 blocks. The color legend lists out the colors and the subspace in Figure 5(e). In figure 5(b) pixels between MBBs 1-3 and MBBs 9-11 (see blocks at position (1,9),(1,10)...(3,11)) are highlighted by brown color which depicts these pair of MBBs overlap maximally in subspace $\{15\}$. The figure also depicts that MBB 6 and MBB 10 do not overlap in any subspace since highlighted by white color. Points within MBBs share maximal k NN relationship in higher order subspaces as the diagonal blocks are highlighted by color depicting the higher order subspaces as listed in legend. In figure 5(d) MBB-4 and MBB-5 maximally overlap in subspace $\{0\}$. Block(5,2) in figure 5(c) have pixel highlights with same color so we capture the same information in minimalistic Heidi in figure 5(d) (see block (5,2)). We do not highlight the whole block with that color since there were overlap in subspace $\{17\}$ only. It also depicts that all points in MBB-5 share k NN relationship with few points in MBB-2 in subspace $\{17\}$ since the pixels were highlighted between all points of MBB-5 (along row) to some points of MBB-2 (along column).

As we observe in minimalistic image block(i, j) and block(j, i) are not necessarily highlighted with same color. The image is not symmetric because kNN relationship is not a symmetric property between a pair of points. So points in $MBB-i$ may share maximal kNN relationship with points in $MBB-j$ in some other subspace as points in $MBB-j$ share to points in $MBB-i$.

4 Experiments

4.1 Synthetic Data Sets

We have generated various synthetic data sets using [14]. We have tested our experiments for $n=100 \dots 5000$ points and $d=2,3,5 \dots 100$ and all such various combinations.

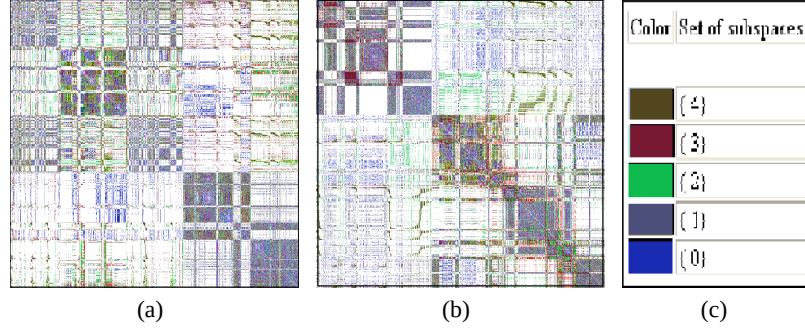


Fig. 6. R-Tree and R*-Tree Heidi Visualization over 5-d data set: (a) Heidi Visualization of R-Tree; (b) Heidi Visualization of R*-Tree; (c) Heidi Color Legend

A dataset having 1000 points and 5 dimensions was generated using [14]. Figure 6(a) displays the Heidi Visualization of R-Tree for it. It has 18 MBB blocks. Heidi visualization has 18×18 blocks. Between every pair of MBB pixels are high-lighted with different colors depicting the presence that different pair of MBBs overlap in different subspace. Figure 6(b) which displays the Heidi Visualization of R*-Tree for this 5-D data having 14 MBB blocks. In a big block at the left most corner of image almost all the pixels of inter MBBs pairs (see blocks at position (1,2), (3,2), (4,7) etc.) are highlighted within maroon color which denotes almost all points of those MBBs pairs share $kNNs$ relationship in $\{3\}$ subspace. MBBs placed at the centre of Heidi image show overlap with some other MBBs only in subspace $\{2\}$ denoted by light green color.

A dataset having 1500 points and 10 dimensions was generated using [14]. Figure 7(a) displays the Heidi Visualization of R-Tree for it. It has 17 MBB blocks. Most of the MBB pairs have a orange strip pattern which depicts points within this MBBs share kNN relationship in $\{9\}$ subspace. There are two big blocks in the Heidi image. They do not show overlap in any of the subspace denoted by white color. Figure 7(b) displays the Heidi Visualization of R*-Tree for this 10-D data. It has 15 MBB blocks. Block (1,4) is highlighted by white colour which depicts points of MBB-1 do not show any overlap with points of MBB-4 in any subspace. For MBBs 2-10 most of the pixels within them highlighted by light green color which denotes points within these MBBs most of the points share kNN relationship in subspace $\{2\}$ with each other. Since R* split over the dimension where minimum perimeter is found and hence it depicts along dimension 2 minimum split was observed. MBB 1 placed at the top left most corner of image do not overlap with any MBB 2-10 since pixels are highlighted by white color.

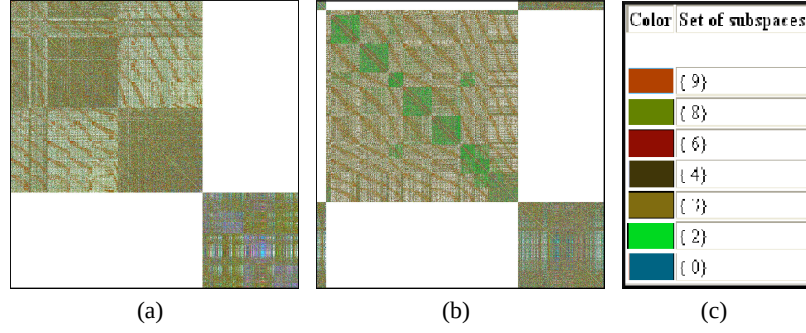


Fig. 7. Heidi Visualization of R-Tree and R*-Tree over 10-d data set:(a) Heidi Visualization of R-Tree;(b) Heidi Visualization of R*-Tree; (c) Heidi Color Legend

4.2 R-Tree and R*-Tree Visualization for Real Life Data Set

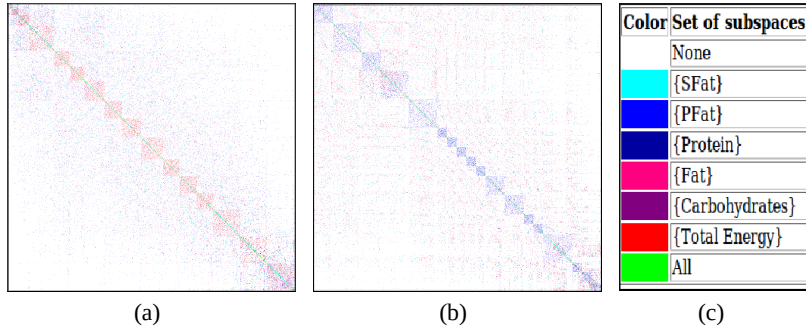


Fig. 8. Heidi Visualization of R-Tree and R*-Tree over Food Data Set:(a) Heidi Visualization of R-Tree; (b) Heidi Visualization of R*-Tree; (c) Heidi Color Legend

We present results of food data set obtained from [2]. The original data set has following dimensions: Id, Name of the food, Total Energy (kcal), Carbohydrates, Fat (g), Protein (g), PFAT (g), SFAT (g). A total of 1493 food items are available in the data set. This is done with purpose to identify foods which share similar value for a dimension(nutrient) and varied values for some other dimension. We can identify foods which can be substitutes for a given food. Heidi Visualization in figure 8 is obtained for single dimensional subspace {Total Energy}, {Carbohydrates}, {Protein}, {Total Fat}, {PFat}, {SFat}.

We construct R-Tree index structure over it and obtain 16 MBBs. The Heidi Visualization of it is shown in Figure 8(a). It is divided into 16 x 16 blocks. Points in the block (MBB-1) at the left most top corner have maximum total energy value whereas points in the block (MBB-16) at the bottom most right corner have minimum energy values. In the visualization blocks corresponding

to MBBs (along the diagonal) have shades of red predominantly; indicating that points within MBBs are close to each other in subspace $\{\text{Total Energy}\}$. Foods in same MBB are having similar energy values. There are some pixel highlighted between MBBs (see blocks at non - diagonal positions). Overlap between MBB-7 and MBB-13 i.e. block(7,13) have maximum pixels highlighted with blue colour which depicts points in MBB-7 share most kNN relationship with points in MBB-13 in $\{\text{Protein}\}$ subspace. And it also depicts there are foods which have similar value for protein but foods in MBB-13 are less energy giving foods as compared to foods in MBB-7. Overlap between MBB-12 and MBB-5 (see block(12,5)) have maximum pixels highlighted with purple shade which depicts pair of points within these two MBBs share most kNN relationship in $\{\text{Carbohydrate}\}$ subspace. And it also infers that there are points (foods) in MBB-5 which has similar values for carbohydrates and are also more energy giving than points (foods) in MBB-12.

We also construct R^* -Tree index structure over it and obtain 18 MBBs. The Heidi Visualization of it shown in Figure 8(b) is divided in into 18×18 blocks. Points(foods) in the block (MBB-1) at the left most top corner have maximum PFAT values whereas points(foods) in the block (MBB-18) at the bottom most right corner have minimum PFAT values. In the visualization blocks corresponding to MBBs (along the diagonal) have blue shades predominantly; indicating that points within MBBs are close to each other in subspace $\{\text{PFAT}\}$. Foods in same MBB are having similar PFAT values. MBB-5 and MBB-11 overlap at block(5,11) have maximum pixels highlighted with red color depicting pair of points within these 2 MBBs share most kNN relationship in $\{\text{Total Energy}\}$ subspace. And foods in MBB-5 has more PFAT values as compared to foods in MBB-11 and are also providing similar total energy values. Overlap between MBB-1 and MBB-9 (see block (1,9)) shows maximum pixels highlighted with dark blue color depicting most kNN relationship in subspace $\{\text{Protein}\}$. Foods in MBB-9 has similar protein values to foods in MBB-1 and are less in PFAT values.

Figure 9 is at MBB level. The size of the dot denotes the extent of the overlap in some subspace. Diagonal blocks have big dots as compared to non - diagonal blocks that depicts points in a MBB itself share most kNN relationship.

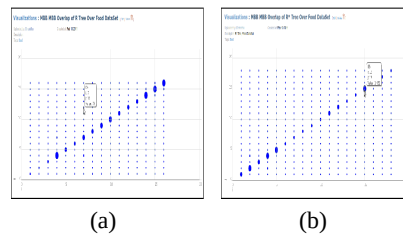


Fig. 9. MBB-MBB Overlap for Food Data Set: (a) MBB-MBB Overlap of R-Tree; (b) MBB-MBB Overlap of R^* -Tree

kNN Query Visualization Over Food DataSet Figure 10 shows an example of a k NN query to find 5 similar foods having similar PFAT value as 'veal' food. Figure 10(a) displays query point along with label (name of the food). User selects the subspace shown in Figure 10(b). Results over R*-Tree index structure and R-Tree index structure are shown with black pixel highlights covered by a bounding box in Figure 10(c) and Figure 10(d) respectively. Similar foods are present in 1 MBB for R*-Tree whereas the query result is spanned over more than 1 MBB for R-Tree index structure. Figure 11 shows an example of a k NN query to find 5 similar foods having similar Total Energy value as 'beef' food. Here similar foods are present in 1 MBB for R-Tree whereas the query result is spanned over more than 1 MBB for R*-Tree index structure. Query results are listed in table 3. We can visualize for first query results were more spanned over more than 1 MBB for R-Tree index structure as compared to R*-Tree index structure where result was spanned in 1 MBB. And for the second query results were spanned over more than 1 MBB for R*-Tree index structure as compared to R-Tree index structure where results was spanned in 1 MBB only. So for first query R*-Tree index structure was better whereas for second query R-Tree index structure was better.

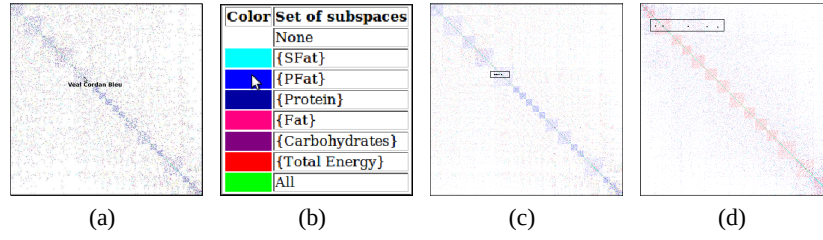


Fig. 10. Explorative Querying for food having similar PFAT value as Veal Corden: (a) Food Selected as Query Point; (b) Selection of Subspace; (c) Results in R*-Tree Index Structure; (d) Results in R-Tree Index Structure

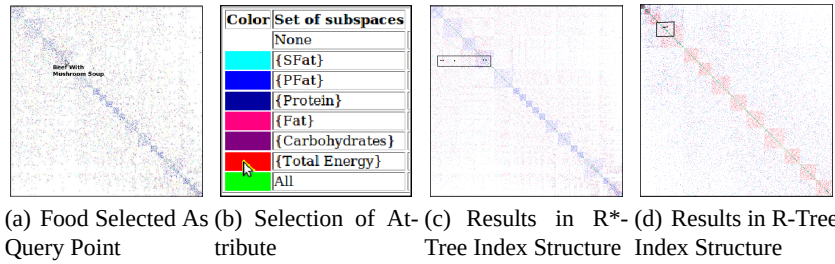


Fig. 11. Explorative Querying for food having similar Energy Value as Beef with Mushroom Soup: (a) Food Selected as Query Point; (b) Selection of Subspace; (c) Results in R*-Tree Index Structure; (d) Results in R-Tree Index Structure

Food Name	k-Value	Subspace(Nutrient)	Similar Foods
Veal cordon bleu	5	PFAT	Veal cordon bleu (input), Chicken or turkey with mushroom soup , Sirloin Chopped with gravy mashed potatoes , Meat loaf made with chicken or turkey, Sardines with tomato-based sauce (mixture) 1 cup
Beef with mushroom soup(1 cup)	5	Total Energy	Beef with mushroom soup(input), Double Cheese Burger (2 patties) , Clam sauce white 1 (cup), Stewed rabbit Puerto Rican style 1 cup NFS, Stewed rabbit Puerto Rican Style 1 cup boneless

Table 3. Queries Over Food Data Set

4.3 Web Interface

A web interface has been developed to display Heidi Visualization of R-Trees for a few data sets generated using Syndeca [14].

More visualizations and results are at:

<http://research.iit.ac.in/~soujanya/heidirtree/>.

We develop R-Tree and R*-Tree index structures for various synthetic data sets [14] and display them. In legend, every subspace is presented as buttons. So if user clicks on any button, points sharing k NN relationship in that subspace are highlighted by that color. The user hovers over the image we present a tool tip of which MBB is the mouse currently pointed to. And if the user click on any MBB we present data analytics listed above.

5 Conclusion

With the example visualizations and experimental results, we demonstrate the following utilities of the visualizing index structures of high dimensional data:

1. Visualization to *understand, analyze* and *study* the R tree and R* tree index structures over a data set.
2. Visualization to *display query* results on high dimensional data
3. Visualization to *evaluate* the index structures based on MBB overlaps and query result visualizations
4. Visualization to bring forth *subspace overlaps* between the MBBs and MBB data analytics to understand the underlying data distribution.

Future work includes building a *R-Tree Diagnostics Tool* which overlays the R-Tree performance statistics with the Heidi Visualization; visually aiding the user to identify frequent page faults and MBB overlaps to amend the structure iteratively. An intuitive interactive user interface needs to be built to realize the concept of “explorative querying” over the data set (performance enhanced with a R-Tree). In future, we intend to parallelize Heidi Matrix computation for better performance and faster visualization.

6 Acknowledgement

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